Nonparametric Filters: review

Probabilistic Robotics, chapter 4

Lesson t2.3 · Thrun, Burgard, Fox

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Six results to hold

1. 4.1 Histogram filter: T p, then multiply and normalize
2. 4.2 Binary Bayes filter: add evidence, subtract the prior
3. 4.3.1 Particle filter: sample, weight, resample
4. 4.3.2 Importance sampling: $w = f/g$, needs $f > 0 \Rightarrow g > 0$
5. 4.3.3 Path space: the weight is $p(z_t \mid x_t)$
6. 4.3.4 Variance, bias, diversity, deprivation

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4.1 Histogram filter (pp. 86–89)

$$p_{k,t} = \eta\, p(z_t \mid x_k) \sum_i p(x_k \mid u_t, x_i)\, p_{i,t-1}$$

Table 4.1

- Predict is T p; update is element-wise, then normalize

Trap: unnormalized columns of T leak mass; normalize per source bin

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4.2 Binary Bayes filter (pp. 94–96)

$$l_t = l_{t-1} + \log \frac{p(x \mid z_t)}{1 - p(x \mid z_t)} - l_0$$

Table 4.2

- Inverse measurement model; evidence is added, prior subtracted

Trap: p. 96 says addition; (4.20) subtracts. Silent at prior 0.5

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4.3.1 Particle filter (pp. 96–100)

1 Line 4

Sample x_t^[m] from the motion model

2 Line 5

Weight $w = p(z_t \mid x_t^{[m]})$

3 Lines 8–11

Resample in proportion to w

- (4.23) holds only as $M \to \infty$; (4.24) carries weights without resampling

Trap: Table 4.3 never normalizes; Table 4.4 needs normalized weights

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4.3.2 Importance sampling (pp. 100–102)

$$w^{[m]} = \frac{f(x^{[m]})}{g(x^{[m]})} \tag{4.27}$$

- Self-normalized (4.28), error O(1/√M); needs f > 0 ⇒ g > 0

Trap: a proposal that misses f returns 0 at every M, no warning

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4.3.3 Path-space derivation (pp. 103–104)

$$w_t^{[m]} = \eta\, p(z_t \mid x_t) \tag{4.34}$$

- Motion term cancels between target and proposal

Trap: the text says Table 4.4 on p. 104; it means Table 4.3

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4.3.4 Variance, bias, deprivation (pp. 105–113)

0.146

variance: std of the mean at M = 25

0.046

the same at M = 250

0.464

bias (4.38): expected mean at M = 2

- M = 1 ignores every measurement; deprivation: no particle near the truth

Trap: injection makes the posterior incorrect; keep off by default

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4.3.4 Resampling (pp. 108–111)

$$i = \operatorname{argmin}_j \sum_{m=1}^{j} w_t^{[m]} \geq U \tag{4.37}$$

- Table 4.4: one random r, teeth U = r + (m − 1)/M, O(M)

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Trap: unnormalized weights send the while loop past the end

C1 Histogram filter, robust

```python
def hist_predict(p, T):
    return T @ p                                    # line 3

def hist_predict_conv(p, kernel):
    # shift-invariant motion on a ring: circular convolution
    return np.real(np.fft.ifft(np.fft.fft(p) * np.fft.fft(kernel)))

def hist_update(log_p_bar, log_lik):
    a = log_lik + log_p_bar                         # line 4, log
    return a - logsumexp(a)                         # eta
```

- Normalize T by column; convolution when motion is shift-invariant
- Update in log space; probe T and the likelihood at the same x̂

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C2 Log-odds update, robust

```python
def logodds_update(l, p_inv, l0, l_min=None, l_max=None):
    l = l + logit(p_inv) - l0       # add evidence, subtract prior
    if l_min is not None:
        l = np.clip(l, l_min, l_max)   # practice, not the book
    return l

def logodds_to_prob(l):
    return expit(l)                 # (4.14), no overflow
```

- Subtract l_0 every update; expit instead of (4.14) as printed
- Clip p before logit; clamp l only as a named choice

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C3 Self-normalized importance sampling

```python
def is_estimate(xs, log_f, log_g, h):
    logw = log_f(xs) - log_g(xs)                # (4.27), log space
    if np.any(np.isposinf(logw)):
        raise ValueError("g = 0 where f > 0: (4.25) violated")
    wn = np.exp(logw - logsumexp(logw))         # (4.28)
    n_eff = 1.0 / np.sum(wn ** 2)
    return float(np.sum(wn * h(xs))), n_eff
```

- Log densities; logsumexp; return N_eff (outside the book)
- Bias of order 1/M from the normalization

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C4 Particle filter step, naive

```
def pf_step_naive(X, u, z, sample_motion, likelihood, rng):
    M = len(X)
    X_bar = np.array([sample_motion(u, X[m], rng)
                      for m in range(M)])            # line 4
    w = np.array([likelihood(z, X_bar[m])
                  for m in range(M)])                # line 5
    idx = rng.choice(M, size=M, p=w / w.sum())       # lines 8-11
    return X_bar[idx]                                # line 12
```

- Robust form: vectorize, log weights, carry weights (4.36)
- Resample on N_eff < M/2 (folklore); wrap angles after line 4

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C5 Table 4.4, zero-based

```
def low_variance_naive(X, w, rng):
    M = len(w)
    out = []
    r = rng.uniform(0, 1 / M)              # line 3
    c = w[0]                               # line 4
    i = 0                                  # line 5
    for m in range(M):                     # line 6
        U = r + m / M                      # line 7
        while U > c:                       # line 8
            i += 1                         # line 9
            c += w[i]                      # line 10
        out.append(X[i])                   # line 12
    return np.array(out)
```

- searchsorted on the cumulative sum is (4.37); clip to M − 1
- Normalize inside the resampler; return indices in real code
- Name the book's stratified apart from the literature's

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4.3.4 Density extraction (pp. 104–106)

One Gaussian

- Weighted mean and covariance (C6)
- Unimodal beliefs only

Multimodal

- k-means mixture
- Histogram: lookup independent of M
- Kernel density

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Trap: report the heaviest cluster's mean, never the mean of all particles

4.1.3 and 4.1.4: skim notes

- (4.6) is exact under (4.2); only $p(z \mid x) \approx p(z \mid \hat{x})$ is approximate
- Line 3's $\bar{p}$ are unnormalized; line 4's η normalizes (p. 92)
- Static vs dynamic decomposition, density trees, selective updating

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What this chapter feeds

- Ch. 8: Table 4.1 becomes grid localization, Table 4.3 becomes MCL
- Ch. 5–6: line 4 is the motion model, line 5 the beam model
- Ch. 9: Table 4.2 once per cell with an inverse sensor model
- Ch. 13: particles carry a map; the path view of 4.3.3 is why

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C6 Weighted mean and covariance

```python
def weighted_mean_cov(X, w, angle_idx=()):
    w = w / w.sum()
    mu = w @ X
    for j in angle_idx:                          # circular mean
        mu[j] = np.arctan2(w @ np.sin(X[:, j]), w @ np.cos(X[:, j]))
    D = X - mu
    for j in angle_idx:
        D[:, j] = wrap(D[:, j])                  # wrap before squaring
    cov = (D * w[:, None]).T @ D / (1.0 - np.sum(w ** 2))
    return mu, cov
```

- Circular mean for θ; divide by $1 - \Sigma\tilde{w}^2$ for unbiased covariance
- One Gaussian only for one mode: cluster first

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Shapes and cost per card

Card	Shapes	Cost
C1 histogram	p (K,), T (K,K)	$O(K^2)$; O(K log K) conv.
C2 log odds	scalar per cell	O(1) per cell
C3 IS	xs (M,n), w (M,)	O(M) evaluations
C4 PF step	X (M,n), logw (M,)	O(M · beams)
C5 resample	w (M,), idx (M,)	O(M); multinomial O(M log M)
C6 density	X (M,n) to μ, Σ	$O(M\ n^2)$; KDE O(M) per query

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Symptom to fix

Symptom	Cause	Fix
Mean between poses	Multimodal belief	Histogram or PF
Map drifts, no evidence	Prior not subtracted	Subtract l_0
Weights zero or NaN	Underflow	Log weights
N_eff $\ll$ M each update	Proposal too wide	Better proposal
Stationary robot certain	Resampling blind	Suspend resampling
Index out of range	Unnormalized weights	Normalize first
Wrong after kidnap	Deprivation	Injection, ch. 8

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Mistakes already made (REVIEW.md)

Where	Error	Fix
L4, C2, map	Clamp attributed to ch. 9	Common practice, not in the book
L3 4.1.3	η of line 3 claimed from (4.10)	p. 92: line 4's η normalizes
Highlight p. 91	Bias direction of dropping $\|x_k\|$	Favours small bins
T1 answer 6	Kidnap breaks $f > 0 \Rightarrow g > 0$	Holds by construction
Session Fig 2.3e	Rate 1/M credited to p. 112	Page gives no rate
Exercises README	Book exercises said to be tests	Adapted, not reproduced
T2.4 notes	Monte Carlo check claimed	No such check exists
(4.4), C4	Flat vs affine; interval	Affine; $[-\pi, \pi)$

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